



Faculty of Computer Science and Mathematics
Universiti Malaysia Terengganu

LAB REPORT

CSE3023 Web-Based Application Development

Name	Wai Pei Yuen
Matric No.	S74991
Semester	2025/26-2
Lab Report	7
Demonstrator	Ts. Dr. Wan Nural Jawahir Hj Wan Yussof

Exercise:

Database

Table registered_subjects

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 	int(100)			No	None		AUTO_INCREMENT	 Change  Drop  More
☐	2 matric_no	varchar(20)	utf8mb4_general_ci		No	None			 Change  Drop  More
☐	3 subject_code	varchar(20)	utf8mb4_general_ci		No	None			 Change  Drop  More
☐	4 subject_name	varchar(100)	utf8mb4_general_ci		No	None			 Change  Drop  More

Source Code:

register.html

```
<!DOCTYPE html>
<!--
Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this template
-->
<html>
  <head>
    <title>Student Registration</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
  </head>
  <body>
    <h2>Register New Student</h2>
    <form action="UserServlet" method="POST" enctype="multipart/form-data">
      <input type="hidden" name="action" value="register">

      Matric No: <input type="text" name="matricNo" required><br><br>
      Full Name: <input type="text" name="fullname" required><br><br>
      Password: <input type="password" name="password" required><br><br>
      Profile Picture: <input type="file" name="profileImage" accept="image/*" required>
      <br>
      <br>
      <input type="submit" value="Register">
    </form>
  </body>
</html>
```

login.html

```
<!DOCTYPE html>
<!--
Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this template
-->
<html>
  <head>
    <title>Login Page</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
  </head>
  <body>
    <h2>Student Login</h2>
    <form action="UserServlet" method="POST">
      <input type="hidden" name="action" value="login">
      Matric No: <input type="text" name="matricNo" required><br><br>
      Password: <input type="password" name="password" required><br><br>
      <input type="submit" value="Login">
    </form>
  </body>
</html>
```

dashboard.jsp

```
<!--
Document   : dashboard
Created on : May 19, 2026, 4:18:05 PM
Author    : wpy92
-->

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="com.lab.bean.StudentBean"%>
<!DOCTYPE html>
<html lang="en">
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<title>Student Dashboard</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
<style>
/* Custom CSS minor untuk saiz dan bingkai gambar sahaja */
.profile-img {
width: 160px;
height: 160px;
object-fit: cover;
border: 5px solid #198754; /* Warna hijau success Bootstrap */
}
</style>
</head>
<body class="bg-light">
<%
// 1. Session Verification (Security Check)
response.setContentType("text/html; charset=UTF-8");
StudentBean user = (StudentBean) session.getAttribute("loggedUser");
if (user == null) {
response.sendRedirect("login.html");
return;
}
%>

<div class="container d-flex justify-content-center align-items-center" style="min-height: 100vh;">
<div class="card shadow-lg border-0 rounded-4 p-4 p-md-5 text-center" style="max-width: 600px; width: 100%;">
<div class="card-body">

<h2 class="fw-bold text-dark mb-2">Welcome, <%= user.getFullname() %>!</h2>
<p class="text-muted fs-5 mb-4">Matric Number: <strong class="text-secondary"><%= user.getMatricNo() %></strong></p>
<div class="d-flex justify-content-center gap-3 flex-wrap">
<a href="subject/viewsSubjects.jsp" class="btn btn-primary px-4 py-2 fw-semibold">Manage Subjects</a>
<a href="UserServlet?action=logout" class="btn btn-dark px-4 py-2 fw-semibold">Logout</a>
<a href="UserServlet?action=delete"
onclick="return confirm('WARNING: Are you sure you want to delete your account permanently?');"
class="btn btn-danger px-4 py-2 fw-semibold">Delete Account
</a>
</div>
</div>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

StudentBean.java

```
package com.lab.bean;

/**
 *
 * @author wpy92
 */
public class StudentBean implements java.io.Serializable {

    private String matricNo;
    private String fullname;
    private String password;
    private String base64Image;

    public StudentBean() {
    }

    public String getMatricNo() {
        return matricNo;
    }

    public void setMatricNo(String matricNo) {
        this.matricNo = matricNo;
    }

    public String getFullname() {
        return fullname;
    }

    public void setFullname(String fullname) {
        this.fullname = fullname;
    }

    public String getPassword() {
        return password;
    }

    public void setPassword(String password) {
        this.password = password;
    }

    public String getBase64Image() {
        return base64Image;
    }

    public void setBase64Image(String base64Image) {
        this.base64Image = base64Image;
    }
}
```

SubjectBean.java

```
package com.lab.bean;

/**
 *
 * @author wpy92
 */
public class SubjectBean implements java.io.Serializable {

    // Define variables
    private int id;
    private String matricNo;
    private String subjectCode;
    private String subjectName;

    // Default constructor
    public SubjectBean() {
    }

    // Getters and setters
    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getMatricNo() {
        return matricNo;
    }

    public void setMatricNo(String matricNo) {
        this.matricNo = matricNo;
    }

    public String getSubjectCode() {
        return subjectCode;
    }

    public void setSubjectCode(String subjectCode) {
        this.subjectCode = subjectCode;
    }

    public String getSubjectName() {
        return subjectName;
    }

    public void setSubjectName(String subjectName) {
        this.subjectName = subjectName;
    }
}
```

StudentDAO.java

```
package com.lab.dao;

/**
 *
 * @author wpy92
 */
import com.lab.bean.StudentBean;
import java.io.ByteArrayOutputStream;
import java.io.InputStream;
import java.sql.Blob;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Base64;

public class StudentDAO {

    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection("jdbc:mysql://localhost:3307/lab7_db", "root", "");
    }

    //--- REGISTER LOGIC---
    public boolean registerStudent(StudentBean student, InputStream imageStream) {
        try (Connection conn = getConnection()) {
            String sql = "INSERT INTO students (matric_no, password, fullname, profile_image) VALUES(?, ?, ?, ?)";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setString(1, student.getMatricNo());
            pstmt.setString(2, student.getPassword());
            pstmt.setString(3, student.getFullname());
            pstmt.setBlob(4, imageStream); // Set BLOB data
            return pstmt.executeUpdate() > 0;
        } catch (Exception e) {
            e.printStackTrace();
            return false;
        }
    }

    //--- LOGIN LOGIC---
    public StudentBean loginStudent(String matricNo, String password) {
        StudentBean student = null;
        try (Connection conn = getConnection()) {
            String sql = "SELECT * FROM students WHERE matric_no = ? AND password = ?";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setString(1, matricNo);
            pstmt.setString(2, password);
            ResultSet rs = pstmt.executeQuery();
            if (rs.next()) {
                student = new StudentBean();
                student.setMatricNo(rs.getString("matric_no"));
                student.setFullname(rs.getString("fullname"));

                // Fetch BLOB and convert to Base64 String
                Blob blob = rs.getBlob("profile_image");
                if (blob != null) {
                    InputStream inputStream = blob.getBinaryStream();
                    ByteArrayOutputStream outputStream = new ByteArrayOutputStream();
                    byte[] buffer = new byte[4096];
                    int bytesRead = -1;
                    while ((bytesRead = inputStream.read(buffer)) != -1) {
                        outputStream.write(buffer, 0, bytesRead);
                    }
                    byte[] imageBytes = outputStream.toByteArray();
                    student.setBase64Image(Base64.getEncoder().encodeToString(imageBytes));
                }
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        return student;
    }
}
```

```
//--- DELETE LOGIC---
public boolean deleteStudent(String matricNo) {
    try (Connection conn = getConnection()) {
        String sql = "DELETE FROM students WHERE matric_no = ?";
        PreparedStatement pstmt = conn.prepareStatement(sql);
        pstmt.setString(1, matricNo);
        return pstmt.executeUpdate() > 0;
    } catch (Exception e) {
        e.printStackTrace();
        return false;
    }
}
```


SubjectDAO.java

```
package com.lab.dao;

/**
 *
 * @author wpy92
 */

import com.lab.bean.SubjectBean;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.ArrayList;
import java.util.List;

public class SubjectDAO {

    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection("jdbc:mysql://localhost:3307/lab7_db", "root", "");
    }

    public boolean addSubject(SubjectBean subject) {
        try (Connection conn = getConnection()) {
            String sql = "INSERT INTO registered_subjects (matric_no, subject_code, subject_name) VALUES(?, ?, ?)";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setString(1, subject.getMatricNo());
            pstmt.setString(2, subject.getSubjectCode());
            pstmt.setString(3, subject.getSubjectName());
            return pstmt.executeUpdate() > 0;
        } catch (Exception e) {
            e.printStackTrace();
            return false;
        }
    }

    public List<SubjectBean> getAllSubjects(String matricNo) {
        List<SubjectBean> list = new ArrayList<>();
        try (Connection conn = getConnection()) {
            String sql = "SELECT * FROM registered_subjects WHERE matric_no = ?";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setString(1, matricNo);
            ResultSet rs = pstmt.executeQuery();
            while (rs.next()) {
                SubjectBean subject = new SubjectBean();
                subject.setId(rs.getInt("id"));
                subject.setMatricNo(rs.getString("matric_no"));
                subject.setSubjectCode(rs.getString("subject_code"));
                subject.setSubjectName(rs.getString("subject_name"));
                list.add(subject);
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        return list;
    }

    public SubjectBean getSubject(int id) {
        SubjectBean subject = null;
        try (Connection conn = getConnection()) {
            String sql = "SELECT * FROM registered_subjects WHERE id=?";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setInt(1, id);
            ResultSet rs = pstmt.executeQuery();
            if (rs.next()) {
                subject = new SubjectBean();
                subject.setId(rs.getInt("id"));
                subject.setMatricNo(rs.getString("matric_no"));
                subject.setSubjectCode(rs.getString("subject_code"));
                subject.setSubjectName(rs.getString("subject_name"));
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        return subject;
    }
}
```

```

public boolean updateSubject(SubjectBean subject) {
    try (Connection conn = getConnection()) {
        String sql = "UPDATE registered_subjects SET subject_code=?, subject_name=? WHERE id=?";
        PreparedStatement pstmt = conn.prepareStatement(sql);
        pstmt.setString(1, subject.getSubjectCode());
        pstmt.setString(2, subject.getSubjectName());
        pstmt.setInt(3, subject.getId());
        return pstmt.executeUpdate() > 0;
    } catch (Exception e) {
        e.printStackTrace();
        return false;
    }
}

public boolean deleteSubject(int id) {
    try (Connection conn = getConnection()) {
        String sql = "DELETE FROM registered_subjects WHERE id = ?";
        PreparedStatement pstmt = conn.prepareStatement(sql);
        pstmt.setInt(1, id);
        return pstmt.executeUpdate() > 0;
    } catch (Exception e) {
        e.printStackTrace();
        return false;
    }
}
}

```

UserServlet.java

```
package com.lab.controller;

import com.lab.bean.StudentBean;
import com.lab.dao.StudentDAO;
import java.io.IOException;
import java.io.InputStream;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.MultipartConfig;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;
import javax.servlet.http.Part;

/**
 *
 * @author wpy92
 */
@MultipartConfig(maxFileSize = 16177215) // max 16 MB required for image upload
public class UserServlet extends HttpServlet {

    private StudentDAO studentDAO = new StudentDAO();

    /**
     * Processes requests for both HTTP GET and POST
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet UserServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet UserServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
     * Handles the HTTP GET method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String action = request.getParameter("action");
        HttpSession session = request.getSession(false);
        if ("logout".equals(action)) {

            /*--- 3. HANDLE LOGOUT---
            if (session != null) {
                session.invalidate(); // Destroy session
            }
            response.sendRedirect("login.html");
            */
        }
    }
}
```

```

else if ("delete".equals(action)) {

    /** 4. HANDLE DELETE ACCOUNT--
    if (session != null && session.getAttribute("loggedUser") != null) {
        StudentBean user = (StudentBean) session.getAttribute("loggedUser");
        studentDAO.deleteStudent(user.getMatricNo());
        session.invalidate(); // Destroy session after account deletion
    }
    response.sendRedirect("register.html");
}

}

/**
 * Handles the HTTP <code>POST</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {

    response.setContentType("text/html;charset=UTF-8");

    // Read the hidden parameter to decide which action to execute
    String action = request.getParameter("action");

    if ("register".equals(action)) {
        /** 1. HANDLE REGISTRATION--
        StudentBean newStudent = new StudentBean();
        newStudent.setMatricNo(request.getParameter("matricNo"));
        newStudent.setFullname(request.getParameter("fullname"));
        newStudent.setPassword(request.getParameter("password"));
        InputStream inputStream = null;
        Part filePart = request.getPart("profileImage");

        if (filePart != null && filePart.getSize() > 0) {
            inputStream = filePart.getInputStream();
        }

        boolean isSuccess = studentDAO.registerStudent(newStudent, inputStream);

        if (isSuccess) {
            response.getWriter().println("Registration Successful! <a href = 'login.html'>Login here</a>");
        }

        else {
            response.getWriter().println("Registration Failed!");
        }
    }

    else if ("login".equals(action)) {
        /** 2. HANDLE LOGIN--
        String matric = request.getParameter("matricNo");
        String pass = request.getParameter("password");
        StudentBean student = studentDAO.loginStudent(matric, pass);

        if (student != null) {
            // Login successful: Store the entire bean in the Session
            HttpSession session = request.getSession();
            session.setAttribute("loggedUser", student);
            response.sendRedirect("dashboard.jsp");
        }

        else {
            response.getWriter().println("Invalid Credentials!<a href = 'login.html'>Try Again</a>");
        }
    }
}

/**
 * Returns a short description of the servlet.
 *
 * @return a String containing servlet description
 */
@Override
public String getServletInfo() {
    return "Short description";
}
}

```

SubjectServlet.java

```
package com.lab.controller;

import com.lab.bean.StudentBean;
import com.lab.bean.SubjectBean;
import com.lab.dao.SubjectDAO;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.RequestDispatcher;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

/**
 *
 * @author wpy92
 */
public class SubjectServlet extends HttpServlet {

    private SubjectDAO subjectDAO = new SubjectDAO();

    /**
     * Processes requests for both HTTP GET and POST
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet SubjectServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet SubjectServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
     * Handles the HTTP GET method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        HttpSession session = request.getSession(false);
        if (session == null || session.getAttribute("loggedUser") == null) {
            response.sendRedirect("login.html");
            return;
        }

        String action = request.getParameter("action");
```

```

        if ("delete".equals(action)) {
            int id = Integer.parseInt(request.getParameter("id"));
            subjectDAO.deleteSubject(id);
            response.sendRedirect("subject/viewSubjects.jsp");
        } else if ("edit".equals(action)) {
            int id = Integer.parseInt(request.getParameter("id"));
            SubjectBean subject = subjectDAO.getSubject(id);
            request.setAttribute("subject", subject);
            RequestDispatcher rd = request.getRequestDispatcher("/subject/updateSubject.jsp");
            rd.forward(request, response);
        }
    }
}

```

```

/**
 * Handles the HTTP <code>POST</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    HttpSession session = request.getSession(false);
    if (session == null || session.getAttribute("loggedUser") == null) {
        response.sendRedirect("login.html");
        return;
    }

    // Retrieve identity directly from Session as per instructions
    StudentBean user = (StudentBean) session.getAttribute("loggedUser");
    String action = request.getParameter("action");
}

```

```

        if ("add".equals(action)) {
            SubjectBean subject = new SubjectBean();
            subject.setMatricNo(user.getMatricNo()); // Identity enforcement
            subject.setSubjectCode(request.getParameter("subjectCode"));
            subject.setSubjectName(request.getParameter("subjectName"));
            subjectDAO.addSubject(subject);
            response.sendRedirect("subject/viewSubjects.jsp");
        }
        else if ("update".equals(action)) {
            SubjectBean subject = new SubjectBean();
            subject.setId(Integer.parseInt(request.getParameter("id")));
            subject.setSubjectCode(request.getParameter("subjectCode"));
            subject.setSubjectName(request.getParameter("subjectName"));
            subjectDAO.updateSubject(subject);
            response.sendRedirect("subject/viewSubjects.jsp");
        }
    }
}

```

```

/**
 * Returns a short description of the servlet.
 *
 * @return a String containing servlet description
 */
@Override
public String getServletInfo() {
    return "Short description";
} // </editor-fold>
}

```

registerSubject.jsp

```
<%--
Document    : registerSubject
Created on  : May 20, 2026, 10:20:31 PM
Author      : wpy92
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    <title>Register Subject</title>
  </head>
  <body>
    <h2>Register Subject</h2>
    <form action="../SubjectServlet" method="POST">
      <input type="hidden" name="action" value="add">

      Subject Code: <input type="text" name="subjectCode" required><br><br>
      Subject Name: <input type="text" name="subjectName" required><br><br>

      <input type="submit" value="Register">
    </form>
    <br>
    <a href="viewSubjects.jsp">View My Subjects</a> | <a href="../dashboard.jsp">Back to Dashboard</a>
  </body>
</html>
```

updateSubject.jsp

```
<%--
Document      : viewSubjects
Created on    : May 20, 2026, 10:30:45 PM
Author       : wpy92
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="java.util.List"%>
<%@page import="com.lab.dao.SubjectDAO"%>
<%@page import="com.lab.bean.SubjectBean"%>
<%@page import="com.lab.bean.StudentBean"%>

<%
// Session Security & Identity Checking
StudentBean user = (StudentBean) session.getAttribute("loggedUser");
if (user == null) {
    response.sendRedirect("../login.html");
    return;
}

SubjectDAO dao = new SubjectDAO();
List<SubjectBean> list = dao.getAllSubjects(user.getMatricNo());
%>

<!DOCTYPE html>
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<title>View Subjects</title>
</head>
<body>
<h2>My Registered Subjects</h2>
<a href="registerSubject.jsp">Add New Subject</a> | <a href="../dashboard.jsp">Back to Dashboard</a>
<br><br>

<table border="1" cellpadding="5">
<tr>
<th>ID</th>
<th>Subject Code</th>
<th>Subject Name</th>
<th>Action</th>
</tr>

<%
// Loop using standard Java Scriptlets
for (SubjectBean s : list) {
<tr>
<td><%= s.getId() %></td>
<td><%= s.getSubjectCode() %></td>
<td><%= s.getSubjectName() %></td>
<td>
<a href="../SubjectServlet?action=edit&id=<%=s.getId()%>">Edit</a> |
<a href="../SubjectServlet?action=delete&id=<%=s.getId()%>" onclick="return confirm('Delete this subject?');">Delete</a>
</td>
</tr>
<%
}
%>
</table>
</body>
</html>
```


viewSubjects.jsp

```
<%--
Document    : updateSubject
Created on  : May 20, 2026, 10:52:56 AM
Author     : wpy92
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="com.lab.bean.SubjectBean"%>

<%
// Retrieve the specific subject data passed from the Servlet
SubjectBean subject = (SubjectBean) request.getAttribute("subject");
if (subject == null) {
    response.sendRedirect("subject/viewSubjects.jsp");
    return;
}
%>

<!DOCTYPE html>
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<title>Update Subject</title>
</head>
<body>
<h2>Update Subject Details</h2>
<form action="SubjectServlet" method="POST">
<input type="hidden" name="action" value="update">
<input type="hidden" name="id" value="<%= subject.getId() %%">

Subject Code: <input type="text" name="subjectCode" value="<%= subject.getSubjectCode() %%" required><br><br>
Subject Name: <input type="text" name="subjectName" value="<%= subject.getSubjectName() %%" required><br><br>

<input type="submit" value="Update">
</form>
<br>
<a href="subject/viewSubjects.jsp">Cancel</a>
</body>
</html>
```

Output:

register.html

Register New Student

Matric No:

Full Name:

Password:

Profile Picture: No file chosen

login.html

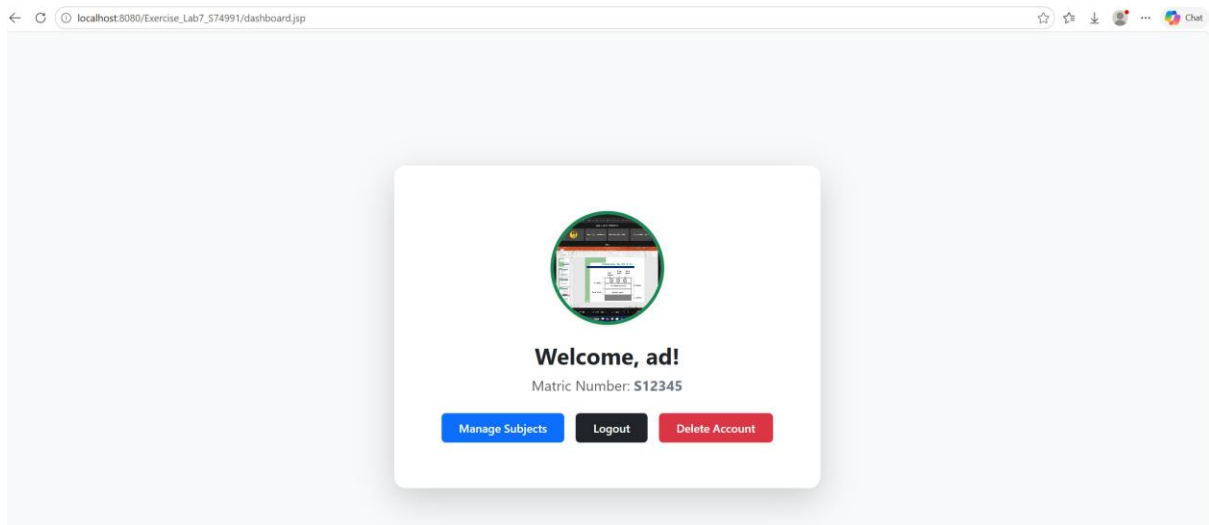
Student Login

Matric No:

Password:

Login

dashboard.jsp



CRUD

View

My Registered Subjects

[Add New Subject](#) | [Back to Dashboard](#)

ID	Subject Code	Subject Name	Action
1	CSF3023	PROGRAMMING	Edit Delete

Create

Register Subject

Subject Code:

Subject Name:

[View My Subjects](#) | [Back to Dashboard](#)

My Registered Subjects

[Add New Subject](#) | [Back to Dashboard](#)

ID	Subject Code	Subject Name	Action
1	CSF3023	PROGRAMMING	Edit Delete
2	CSE3433	ARCHITECTURE	Edit Delete

← T →				id	matric_no	subject_code	subject_name
☐	 Edit	 Copy	 Delete	1	S12345	CSF3023	PROGRAMMING
☐	 Edit	 Copy	 Delete	2	S12345	CSE3433	ARCHITECTURE

Update

Update Subject Details

Subject Code:

Subject Name:

[Cancel](#)

Update Subject Details

Subject Code:

Subject Name:

[Cancel](#)

My Registered Subjects

[Add New Subject](#) | [Back to Dashboard](#)

ID	Subject Code	Subject Name	Action
1	CSF3023	OOP	Edit Delete
2	CSE3433	ARCHITECTURE	Edit Delete

← T → ▼ id matric_no subject_code subject_name							
☐	 Edit	 Copy	 Delete	1	S12345	CSF3023	OOP
☐	 Edit	 Copy	 Delete	2	S12345	CSE3433	ARCHITECTURE

Delete

← ↻ ⓘ localhost:8080/Exercise_Lab7_S74991/subject/viewSubjects.jsp

My Registered Subjects

[Add New Subject](#) | [Back to Dashboard](#)

ID	Subject Code	Subject Name	Action
1	CSF3023	OOP	Edit Delete
2	CSE3433	ARCHITECTURE	Edit Delete

localhost:8080 says

Delete this subject?

OK

Cancel

My Registered Subjects

[Add New Subject](#) | [Back to Dashboard](#)

ID	Subject Code	Subject Name	Action
1	CSF3023	OOP	Edit Delete

← T →	▼	id	matric_no	subject_code	subject_name
☐	 Edit	 Copy	 Delete	1	S12345
				CSF3023	OOP